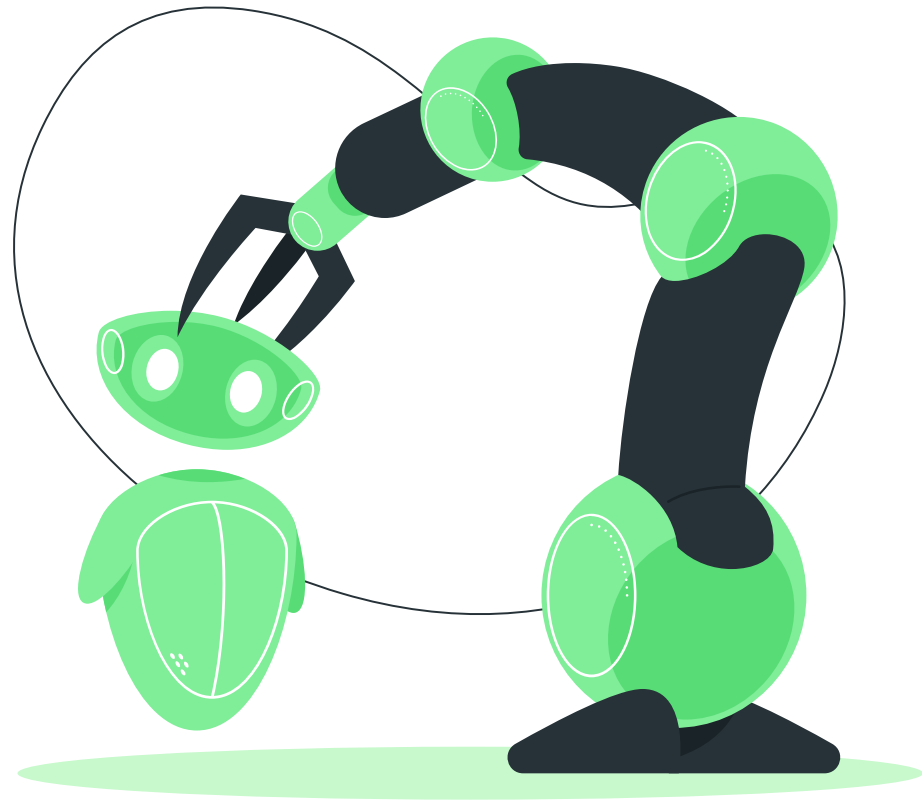


Artificial Intelligence

Driven Robotics

Chris Felumero



Presentation Overview

~Background

~How AI is being used

~Artifact

~Conclusion

~Topic Overview

~Mentoring

~Capstone Experience

~Questions

Background

About Me

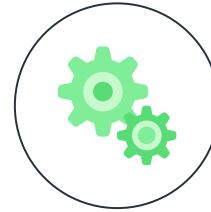
- Programming Lead for robotics team
- Computer Science

Concepts



Artificial Intelligence (AI)

the simulation of human intelligence in machines that are programmed to think like humans and mimic their actions.

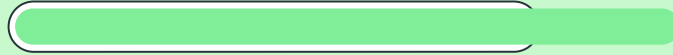


AI-Driven Robotics

the field of robotics that uses artificial intelligence and machine learning to build and program robots to automate or carry out tasks

Why are companies increasing the usage of AI Robots?

120
%



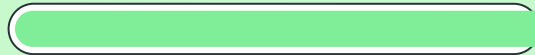
Human labor cost
increase since
1990 (EIU, 2017)

40%



Robot cost
decrease since
1990 (ECU, 2017)

484%



Projected increase in
the AI-Robotics
market in the next 8
years (PR, 2022)

Industries using AI-Driven Robotics

1

Shipping

3

Transportation

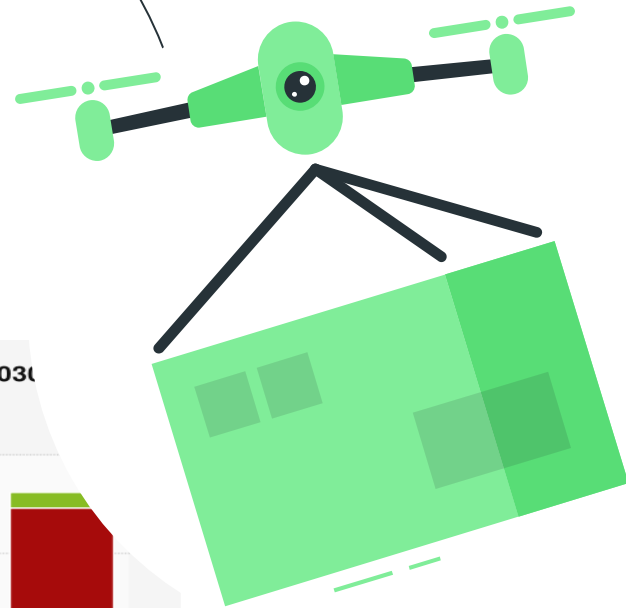
2

Manufacturing

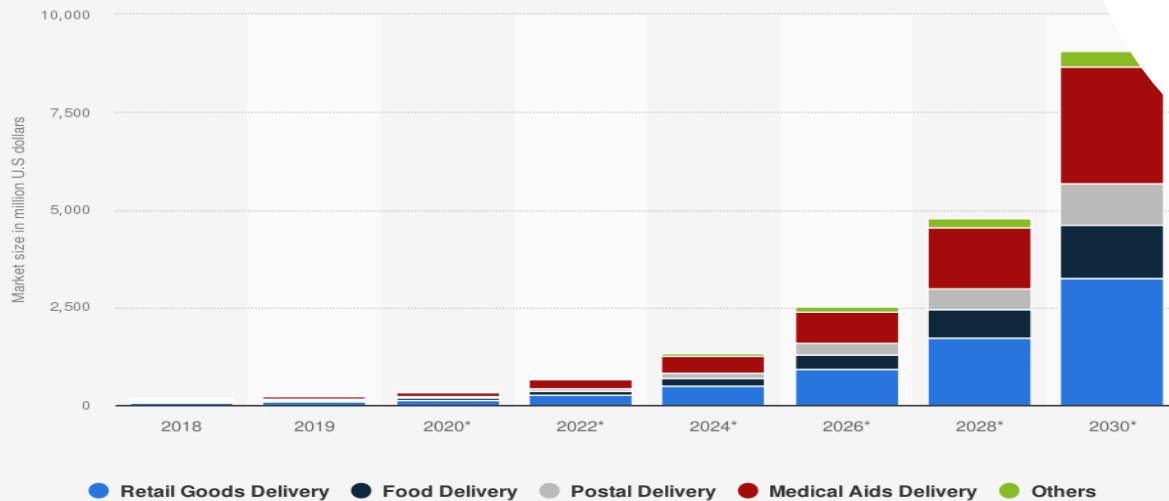
4

Construction

Shipping



**Drone package delivery system market size in the United States from 2018 to 2030
end use (in million U.S. dollars)**



Construction

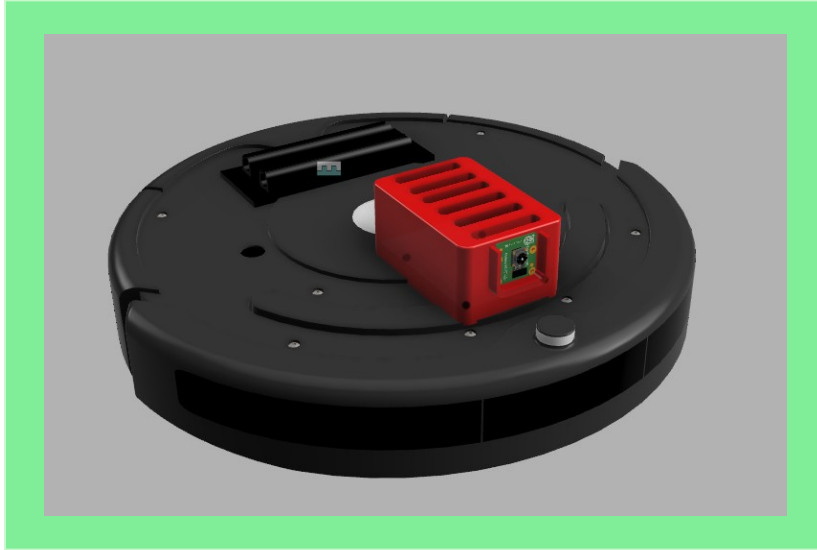


Mentoring

Thank you to my mentor, Kevin Monteith



Artifact



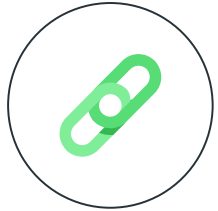
(T)racking
(R)obot using
(A)rtificial
(I)ntelligence
(N)avigation

Features of TRAIN



Autonomous Navigation

Uses machine learning to navigate towards an object autonomously



AI Object Tracking

Uses object recognition software to locate and track an object



iRobot Create 2

A modified version of a Roomba used as a base for TRAIN



Raspberry Pi 3

Micro-Computer used to program and operate TRAIN

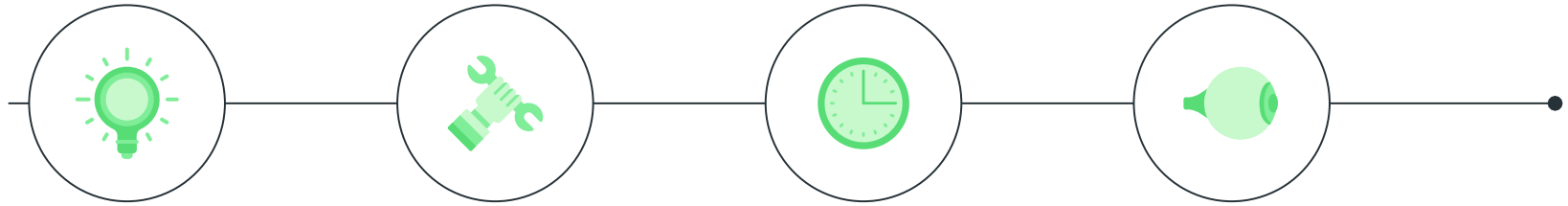
Product Development Timeline

Design

Completed 11/22/2022

Programming

Completed 12/28/2022



Prototyping

Completed 12/7/2022

Product Completed

Completed 12/29/2022

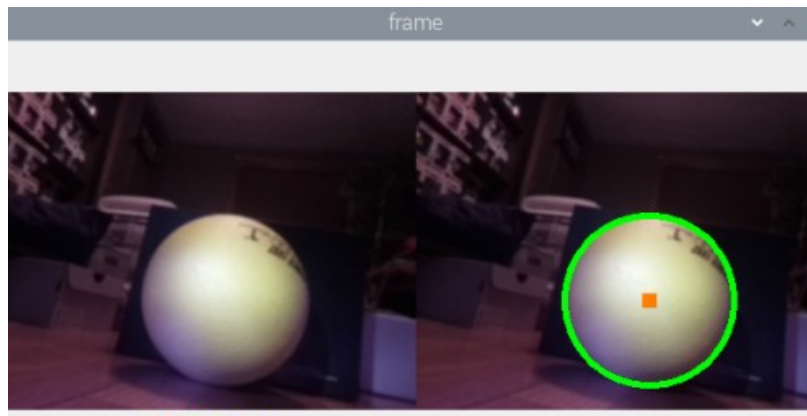
TRAIN Video



How Train Works

-Object Tracking

-Bounding Box



Capstone Experience

- Public speaking
- Research
- Time management

Conclusion

With all of the improvements in technology and the increased use of AI driven robotics across many industries, society's workforce is undergoing a major change in how it operates. Within the next few years, it will become more and more common for humans to share work environments with robots and rely on robots for labor.

Questions

Are there any questions?

